Chapter III.9

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II.3.17 *Zariski Spaces.* A topological space X is a *Zariski space* if it is noetherian and every (nonempty) closed irreducible subset of X has a unique generic point.

(a) Show that if X is a noetherian scheme, then $\mathrm{Sp}(X)$ is a Zariski space.

(b) show that any minimal nonempty closed subset of a Zariski space onsists of one point. We call these *closed points.*

(c) Show that a Zariski space X satisfies the axiom T_0: given any two distinct points of X, there is an open set containing one but not the other.

(d) If X is an irreducible Zariski space, then its generic point is contained in every nonempty open subset of X.

(e) If x_0, x_1 are points of a topological space X ,and if $x_0 \in \{x_1\}^-$, then we say x_1 *specializes* to x_0, written $x_1 \rightsquigarrow x_0$. We also say x_0 is a *specialization* of x_1, or that x_1 is a *generalization of* x_0. Now, let X be a Zariski space. Show that the minimal points, for the partial ordering determined by $x_1 > x_0$ if $x_1 \rightsquigarrow x_0$, are the closed points, and the maximal points are the generic points of the irreducible components of X. Show also that a closed subset contains every specialization of any of its points. (We say closed subsets are *stable under specialization.*) Similarly, open subsets are *stable under generalization.*

(f) Show that the maximal spectrum of a ring A is Zariski if and only if it is equal to $\mathrm{Spec}\, A$.

Proof. (a) Trivial, given (II, Ex. 2.9).

(b) Let X be a Zariski space, and let $\emptyset \neq Y \subset X$ be a minimal nonempty closed subset of X. In particular, Y is irreducible, and thus contains a unique generic point $\zeta \in Y$. Minimality implies $\{y\}^- = Y$ for all $y \in Y$, but then uniqueness implies $Y = \{\zeta\}$, as desired.

(c) Let $x, y \in X$ be points such that $x \in U$ open implies $y \in U$. Then $y \in C$ closed implies $x \in C$, so that $\{y \in C \text{ closed}\} \subset \{x \in C \text{ closed}\}$. Hence,

$$\{x\}^- = \bigcap_{x \in C} C \subset \bigcap_{y \in C} C = \{y\}^-.$$

If $x, y \in X$ have $x \in U$ if and only if $y \in U$, then $\{x\}^- = \{y\}^-$ implies $x = y$.

(d) If $\emptyset \neq U \subset X$ is a nonempty open set, then $x \notin U$ implies $\{x\}^- \subset U^c \subsetneq X$ is proper, and hence x is not generic.

(e) Let $x \in X$ be minimal. $y \in \{x\}^-$ implies $x \rightsquigarrow y$, so that $y = x$. Hence, x is closed. Now, let $\zeta \in X$ be maximal. $\{\zeta\}^- \subset \{y\}^-$ implies $y \rightsquigarrow \zeta$, so that $y = \zeta$. Each irreducible closed subset of X has a generic point by definition, so we have that $\{\zeta\}^-$ is indeed an irreducible component of X.

Let $x \in C$ be contained in a closed set C, and take y such that $x \rightsquigarrow y$. Then $y \in \{x\}^- \subset C$, as desired. Now, we claim that $S \in X$ is closed under generalization if and only if S^c is closed under specialization. If $x \in S$ and $y \rightsquigarrow x$, then $y \in S^c$ and S^c closed under specialization

would imply $x \in S^c$, a contradiction. The converse is identical, and our result for open sets follows.

(f) One direction is clear. □

II.3.18 *Constructible Sets.* Let X be a Zariski topological space. A *constructible subset* is a subset which belongs to the smallest family $\mathfrak{F}$ such that (1) every open subset is in $\mathfrak{F}$, (2) a finite intersection of elements of $\mathfrak{F}$ is in $\mathfrak{F}$, and (3) the complement of any element of $\mathfrak{F}$ is in $\mathfrak{F}$.

(a) A subset of X is *locally closed* if it is the intersection of an open subset with a closed subset. Show that a subset of X is constructible if and only if it can be written a finite disjoint union of locally closed subsets.

(b) Show that a constructible subset of an irreducible zariski space X is dense if and only if it contains the generic point. Furthermore, in that case it contains a nonempty open subset.

(c) A subset S of X is closed if and only if it is constructible and stable under specialization. Similarly, a subset T of X is open if and only if it is constructible and stable under generalization.

(d) If $f : X \to Y$ is a continuous map of Zariski spaces, then the inverse image of any constructible set of Y is constructible in X.

Proof. (a) Write $\mathfrak{L}$ for the family of finite disjoint unions of locally closed subsets. It is clear that $\mathfrak{L} \subset \mathfrak{F}$, so it suffices to show that $\mathfrak{F} \subset \mathfrak{L}$. We will do so by showing $\mathfrak{L}$ satisfies (1-3), and conclude our result via minimality.

(1) is trivial. If $X = X_1 \cap X_2$ for $X_i \in \mathfrak{L}$, then we can write

$$X_i = U_{i1} \cap C_{i1} \sqcup \cdots \sqcup U_{im} \cap C_{im} \quad (i = 1, 2),$$

adding null sets if necessary so that each is a union of m locally closed sets. Then

$$\begin{aligned} X &= \left(\bigsqcup_{j=1}^{m} (U_{1j} \cap C_{1j}) \right) \cap \left(\bigsqcup_{j=1}^{m} (U_{2j} \cap C_{2j}) \right) \\ &= \bigsqcup_{1 \leqslant j,k \leqslant m} (U_{1j} \cap U_{2k}) \cap (C_{1j} \cap C_{2k}) \in \mathfrak{L}, \end{aligned}$$

Since the union is clearly disjoint. Arbitrary finite intersections follow by induction, giving us (2).

Suppose

$$X = U_1 \cap C_1 \sqcup \cdots \sqcup U_m \cap C_m \in \mathfrak{L}.$$

Then

$$\begin{aligned} X^c &= U_1 \cap C_1 \sqcup \cdots \sqcup U_m \cap C_m \\ &= (U_1^c \cup C_1^c) \cap \cdots \cap (U_m^c \cup C_m^c) \\ &= ((U_1 \cap C_1^c) \sqcup U_1^c) \cap \cdots \cap ((U_m \cap C_m^c) \sqcup U_m^c) \in \mathfrak{L} \end{aligned}$$

by (2), whence we $\mathfrak{L}$ satisfies (3). We conclude that in fact $\mathfrak{F} = \mathfrak{L}$

(b) Let X be an irreducible Zariski space with generic point ζ, and let $Y \in \mathfrak{F}$ be constructible and dense, i.e. $Y^- = X$. By (a), we can write $Y = \bigsqcup_{i=1}^{n} (U_i \cap C_i) \in \mathfrak{F}$ for $U_i \neq \emptyset$ open and $C_i \neq \emptyset$ closed. Then certainly $X = Y^- \subset \bigcup_{i=1}^{n} C_i$, so X irreducible implies $X = C_i$ for some i, say $X = C_1$. In particular, $\zeta \in C_1$. Further, we have $\zeta \in U_1$, as otherwise

$\zeta \in U_1^c \subsetneq X$ contradicts genericity of ζ. We conclude that, $\zeta \in U_1 = C_1 \cap U_1 \subset Y$, whence both results follow.

(c) (a) and (Ex. 3.17e) give us one direction, while the proof of the latter gives us that S is closed under specialization if and only if S^c is closed under generalization. Now, let $S \in \mathfrak{F}$ be constructible and closed under specialization. It suffices to show that S is closed. Suppose first that $S = U \cap C \neq \emptyset$ for C irreducible with generic point ζ. Then (b) implies $\zeta \in S$, whence $S = C$ by stability under generalization.

Now, take $S = \bigcup_{i=1}^n U \cap C_i$ for C_i irreducible and $U \cap C_i \neq \emptyset$. We claim that each $U \cap C_i$ is closed under specialization. Let $x \in U \cap C_i$ and take $x \rightsquigarrow y$. Then $y \in C_i$ and $y \in U \cap C_j \subset U$ for some j, so, indeed, $y \in U \cap C_i$. It follows that the result holds when $S = U \cap C$ for U open and C closed.

Finally, (a) allows us to write $S = \bigsqcup_{i=1}^n (U_i \cap C_i)$ for U_i open and C_i closed. We claim that each $U_i \cap C_i$ is closed under specialization, whence the result would follow from our previous claim. Let $x \in U_i \cap C_i$ and take $x \rightsquigarrow y$. Then $y \in U_j \cap C_j$ for some j, so closure of open sets under generalization gives us $x \in U_j$, Suppose, by way of contradiction, that $i \neq j$. Disjointness implies $y \in U_i^c$ and $x \in C_j^c$. Then

$$\{y\}^- \subset \{x\}^- \cap C_j \subset \{x\}^- \setminus \{x\}.$$

FINISH THIS

(d) If $S = \bigsqcup_{i=1}^n (U_i \cap C_i)$ is constructible in Y, then

$$\begin{aligned} f^{-1}(S) &= \bigcup_{i=1}^n f^{-1}(U_i \cap C_i) \\ &= \left(\bigcap_{i=1}^n (f^{-1}(U_i) \cap f^{-1}(C_i))^c \right)^c \end{aligned}$$

is constructible in X. □

II.3.19 Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then the image of any constructible subset of X is a constructible subset of Y. Prove this theorem in the following steps.

(a) Reduce to showing that $f(X)$ itself is constructible, in the case where X and Y are affine, integral noetherian schemes, and f is a dominant morphism.

(b) In that case, show that $f(X)$ contains a nonempty open subset of Y by using the following result from commutative algebra: let $A \subset B$ be an inclusion of noetherian integral domains, with B a finitely generated A-algebra. Then, given a nonzero element $b \in B$, there is a nonzero element $a \in A$ with the following property: if $\phi : A \to K$ is any homomorphism of A to any algebraically closed field K, such that $\phi(a) \neq 0$, then ϕ extends to a homomorphism ϕ' of B into K, such that $\phi'(b) \neq 0$. [*hint*: Prove the algebraic result by induction on the number of generators of B over A, with a direct proof for B cyclic over A. In the application, take $b = 1$.]

(c) Now use noetherian induction on Y to complete the proof.

(d) Give some examples of morphisms $f : X \to Y$ of varieties over an algebraically closed field k, to show that $f(X)$ need not be either open or closed.

Proof. (a) □

9.1. A flat morphism $f : X \to Y$ of finite type of noetherian schemes is open, i.e. for every open subset $U \subset X$, $f(U)$ is open in Y. [*Hint*: show that $f(U)$ is constructible and stable under generalization (II, Ex. 3.18) and (II, Ex. 3.19).]

Proof. $f(U)$ is constructible by (II, Ex. 3.19), so it suffices to show it is stable under generalization by (II, Ex. 3.18c). □

9.3. Some examples of flatness and nonflatness.

(a) If $f : X \to Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k, then k is flat.

(b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y. Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x, y]$ and $X = \operatorname{Spec}[x, y, z, w]/(z, w) \cap (x + z, y + w)$.

(c) Again let $Y = \operatorname{Spec} k[x, y]$, but take $X = \operatorname{Spec} k[x, y, z, w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$ has no embedded points, but that f is not flat.

9.4. *Open Nature of Flatness.* Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then $\{x \in X | f \text{ if flat at } x\}$ is an open subset of X (possibly empty).

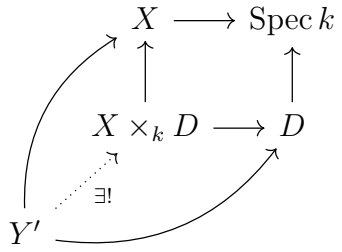
Proof. □

9.5.

9.7. Let $Y \subset X$ be a closed subscheme, where X is a scheme of finite type over a field k. Let $D = k[t]/t^2$ be the ring of dual numbers, and define an *infinitesimal deformation of Y as a closed subscheme of X*, to be a closed subscheme $Y' \subset X \times_k D$, which is flat over D and whose closed fibre is Y. Show that there are classified by $H^0(Y, \mathscr{N}_{Y/X})$, where

$$\mathscr{N}_{Y/X} = \mathcal{H}om_{\mathscr{O}_X}(\mathscr{I}_Y/\mathscr{I}_Y^2, \mathscr{O}_Y).$$

Proof. The universal property of $X \times_k D$ gives us that maps $Y' \to X \times_k D$ are in correspondence with commuting diagrams of the following form.



□

9.10 A scheme X_0 over a field k is *rigid* if it has no infinitesimal deformations.

(a) Show that $\mathbb{P}^1_k$ is rigid, using (9.13.2).